

Boundedness of the Auxiliary Operators in the Generalized Krein–von Neumann Construction

Run `fa_banach_001`, lane 8

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Status and source question

Status: candidate full solution, likely valid; human review requested.

Tarcsay and Titkos consider a subpositive operator $A : \text{dom } A \rightarrow F$ on a weak-star sequentially complete anti-dual pair $\langle F, E \rangle$. Under the hypotheses of their Theorem 3.3, two weakly continuous maps $B, C : E \rightarrow E$ leave $\text{dom } A$ invariant, the spectrum of $BC|_{\text{dom } A}$ is bounded, and

$$C^*A \subset AB, \quad B^*A \subset AC.$$

The source theorem and the definitions of its auxiliary operators are shown below [1].

The next theorem tells us that the Krein–von Neumann extension preserves certain commutation properties as well. We remark in advance that the proof is a bit unusual: although the claim concerns continuous operators, the proof itself applies unbounded operator techniques.

Theorem 3.3. *Let $A : \text{dom } A \rightarrow F$ be a subpositive operator on the w^* -sequentially complete anti-dual pair $\langle F, E \rangle$. Suppose that there are two weakly continuous operators $B, C \in \mathcal{L}(E; E)$ leaving $\text{dom } A$ invariant, and that the spectrum of BC restricted to $\text{dom } A$ is bounded. Assume in addition that B and C satisfy*

$$(3.9) \quad C^*A \subset AB, \quad \text{and} \quad B^*A \subset AC,$$

then the Krein–von Neumann extension of A satisfies

$$C^*A_N = A_NB, \quad \text{and} \quad B^*A_N = A_NC.$$

Proof. Let us introduce two linear operators $\widehat{B}$ and $\widehat{C}$ on the dense linear subspace $\text{ran } A \subseteq H_A$ by letting

$$\widehat{B}(Ax) := ABx, \quad \widehat{C}(Ax) := ACx, \quad x \in \text{dom } A.$$

Observe that $\widehat{B}$ and $\widehat{C}$ satisfy

$$(\widehat{B}(Ax) | Ay)_A = (Ax | \widehat{C}(Ay))_A, \quad x, y \in \text{dom } A$$

because from (3.9) it follows that

$$(ABx | Ay)_A = \langle ABx, y \rangle = \langle Ax, Cy \rangle = (Ax | ACy)_A.$$

We see therefore that $\widehat{B}$ and $\widehat{C}$ are closable operators and their closures satisfy

$$(3.10) \quad (\widehat{B}^{**}h | k)_A = (h | \widehat{C}^{**}k)_A, \quad h \in \text{dom}(\widehat{B}^{**}), k \in \text{dom}(\widehat{C}^{**}).$$

We are going to prove first that $\widehat{B}^{**}$ and $\widehat{C}^{**}$ are adjoints of each other, i.e.,

$$\widehat{C}^* = \widehat{B}^{**}, \quad \widehat{B}^* = \widehat{C}^{**}.$$

By [25, Theorem 3.1] that will be shown once we prove that the operator matrix

$$\Pi := \begin{pmatrix} tI_A & -\widehat{B}^{**} \\ \widehat{C}^{**} & tI_A \end{pmatrix}$$

The auxiliary Hilbert space H_A is the completion of $\text{ran } A$ for

$$(Ax \mid Ay)_A = \langle Ax, y \rangle, \quad x, y \in \text{dom } A,$$

and the densely defined auxiliary operators are

$$\widehat{B}(Ax) = ABx, \quad \widehat{C}(Ax) = ACx. \quad (1)$$

The source proves that their closures are mutual adjoints:

$$\widehat{C}^* = \widehat{B}^{**}, \quad \widehat{B}^* = \widehat{C}^{**}. \quad (2)$$

It then explicitly leaves their boundedness open outside the Banach-space specialization.

$$(3.11) \quad \{(A(t^2I + BC)x, A(t^2I + CB)y) \mid x, y \in \text{dom } A\} \subset H_A \times H_A.$$

By assumption, we have that the restriction $t^2I + BC|_{\text{dom } A} : \text{dom } A \rightarrow \text{dom } A$ is weakly continuously invertible whenever t is large enough. The same holds true for $t^2I + CB$ because of the Jacobson lemma. Hence we conclude that the set in (3.11) is identical with $\text{ran } A \times \text{ran } A$, and therefore $\text{ran } A \times \text{ran } A \subseteq \text{ran } \Pi_0$, which proves our claim.

Next we show that $\text{ran } J^* \subseteq \text{dom}(\widehat{B}^{**})$ and $J^*B = \widehat{B}^{**}J^*$. Let $y \in E$ and $x \in \text{dom } A$, then $(\widehat{C}(Ax) \mid J^*y)_A = \langle ACx, y \rangle = \langle Ax, By \rangle = (Ax \mid J^*By)_A$, whence we have $J^*y \in \text{dom}(\widehat{C}^*)$ and $\widehat{C}^*J^*y = J^*By$. Since we have $\widehat{C}^* = \widehat{B}^{**}$, this proves $J^*B = \widehat{B}^{**}J^*$. Similarly, $J^*C = \widehat{C}J^*$. And now we have arrived:

$$A_N B = J J^* B = J \widehat{B}^{**} J^* = (J \widehat{B}^* J^*)^* = (J \widehat{C}^{**} J^*)^* = (A_N C)^* = C^* A_N,$$

and similarly, $A_N C = B^* A_N$. The proof is complete. $\square$

We have proved above that the auxiliary operators $\widehat{B}$ and $\widehat{C}$ (more precisely, their closures) are adjoint of each other, but we do not know whether they are bounded or not. If the underlying anti-dual pair comes from a Banach space E and its conjugate dual space $F = E'$ then one can imitate the proof of [14 Theorem 2.2] to show that $\widehat{B}, \widehat{C}$ are bounded operators with common norm bound $\sqrt{r(BC)}$, the spectral radius of BC .

Thus the exact source question is whether $\widehat{B}, \widehat{C}$, more precisely their closures, are bounded in the full anti-dual-pair setting. The paragraph also records the expected Banach-space bound $\sqrt{r(BC)}$.

Idea of the proof

Put $T = \widehat{B}^{**}$. The source identity (2) makes $T^* = \widehat{C}^{**}$, so $Q = TT^*$ is a positive self-adjoint operator. On the dense subspace $\text{ran } A$, this product is precisely the operator induced by BC .

If λ lies above the bounded algebraic spectrum of $BC|_{\text{dom } A}$, then $\lambda - BC$ is onto $\text{dom } A$. After applying A , this says

$$\text{ran } A \subseteq \text{ran}(\lambda - Q).$$

For a self-adjoint operator, membership of a vector y in this range makes the inverse-square potential of its spectral measure finite at λ . This happens at *every* real λ above the spectral radius.

The key elementary lemma says that a finite measure on the real line cannot have finite inverse-square potential at every point of an interval unless it vanishes on that interval. Applying it to the spectral measures of a dense set kills the spectral projection of Q above the radius. Thus Q is bounded, which forces T and T^* bounded with the desired square-root bound.

Inverse-square measure lemma

Lemma 1. *Let μ be a finite positive Borel measure on $\mathbb{R}$, and let $I \subseteq \mathbb{R}$ be an open interval. If*

$$\int_{\mathbb{R}} \frac{d\mu(t)}{|t - \lambda|^2} < \infty \quad \text{for every } \lambda \in I, \quad (3)$$

then $\mu(I) = 0$.

Proof. Fix a compact interval $K = [a, b] \subset I$. For each $\lambda \in K$, (3) first implies $\mu(\{\lambda\}) = 0$, and then

$$\frac{\mu((\lambda - h, \lambda + h))}{h^2} \leq \int_{0 < |t - \lambda| < h} \frac{d\mu(t)}{|t - \lambda|^2} \longrightarrow 0 \quad (h \downarrow 0). \quad (4)$$

The convergence follows by absolute continuity of the integral of the integrable function $|t - \lambda|^{-2}$.

For $m \geq 1$, let E_m consist of the points $\lambda \in K$ for which

$$\mu((\lambda - h, \lambda + h)) \leq h^2 \quad (0 < h < 1/m).$$

Equation (4) gives $K = \bigcup_{m \geq 1} E_m$. We show that each E_m has zero outer μ -measure. Fix $0 < \delta < 1/(5m)$. The one-dimensional $5r$ -covering lemma [3, Chapter 2] supplies pairwise disjoint intervals

$$B_j = (\lambda_j - r_j, \lambda_j + r_j), \quad \lambda_j \in E_m, \quad r_j < \delta,$$

such that $E_m \subseteq \bigcup_j 5B_j$. The defining estimate for E_m applied at radius $5r_j$ gives

$$\mu^*(E_m) \leq \sum_j \mu(5B_j) \leq 25 \sum_j r_j^2 \leq 25\delta \sum_j r_j.$$

The disjoint intervals B_j lie in $[a - \delta, b + \delta]$, hence

$$\sum_j r_j \leq \frac{b - a + 2\delta}{2}.$$

Letting $\delta \downarrow 0$ proves $\mu^*(E_m) = 0$. Therefore $\mu(K) = 0$, and exhaustion of I by compact intervals completes the proof. $\square$

Full boundedness theorem

Theorem 1. *Assume all the hypotheses of Theorem 3.3 of Tarcsay–Titkos described above. Let*

$$S = BC|_{\text{dom } A} : \text{dom } A \rightarrow \text{dom } A, \quad \rho = \sup\{|\zeta| : \zeta \in \sigma(S)\} < \infty,$$

where $\rho = 0$ if $\sigma(S) = \emptyset$. Then the two auxiliary operators in (1) have bounded everywhere-defined closures, and

$$\boxed{\|\widehat{B}^{**}\| = \|\widehat{C}^{**}\| \leq \sqrt{\rho}.} \quad (5)$$

Proof. Set

$$D = \text{ran } A \subseteq H_A, \quad T = \widehat{B}^{**}.$$

The subspace D is dense by the construction of H_A . By (2),

$$T^* = \widehat{B}^* = \widehat{C}^{**}.$$

In particular, T is a closed densely defined operator. The standard von Neumann product theorem therefore says that

$$Q := TT^* \tag{6}$$

is a nonnegative self-adjoint operator on H_A [2, Chapter VIII].

Both initial auxiliary operators leave D invariant. Hence $D \subseteq \text{dom } Q$, and for $x \in \text{dom } A$,

$$Q(Ax) = T\widehat{C}(Ax) = \widehat{B}\widehat{C}(Ax) = A(BC)x. \tag{7}$$

No injectivity of A is used here.

Fix a real number $\lambda > \rho$. Then $\lambda I - S : \text{dom } A \rightarrow \text{dom } A$ is bijective by the definition of the algebraic spectrum used in the source; only surjectivity will be needed. Given $y = Az \in D$, choose $x \in \text{dom } A$ such that

$$(\lambda I - BC)x = z.$$

Equation (7) gives

$$y = (\lambda I - Q)Ax. \tag{8}$$

Thus

$$D \subseteq \text{ran}(\lambda I - Q) \quad \text{for every real } \lambda > \rho.$$

Let E_Q be the projection-valued spectral measure of Q . For $y \in H_A$, write

$$\mu_y(\Delta) = \|E_Q(\Delta)y\|^2 \quad (\Delta \subseteq \mathbb{R} \text{ Borel}).$$

If $y \in D$, (8) supplies $u = Ax \in \text{dom } Q$ with $y = (\lambda I - Q)u$. The spectral theorem gives

$$d\mu_y(t) = |\lambda - t|^2 d\mu_u(t).$$

In particular, $\mu_y(\{\lambda\}) = 0$ and

$$\int_{\mathbb{R}} \frac{d\mu_y(t)}{|t - \lambda|^2} = \mu_u(\mathbb{R} \setminus \{\lambda\}) \leq \|u\|^2 < \infty. \tag{9}$$

This is valid for every $\lambda \in (\rho, \infty)$. Lemma 1 therefore yields

$$\mu_y((\rho, \infty)) = 0, \quad y \in D.$$

Equivalently,

$$E_Q((\rho, \infty))y = 0, \quad y \in D.$$

Since D is dense and a spectral projection is bounded,

$$E_Q((\rho, \infty)) = 0. \tag{10}$$

The operator Q is nonnegative. Equation (10) therefore confines its spectral measure to $[0, \rho]$. The spectral theorem now makes Q bounded and everywhere defined, with

$$0 \leq Q \leq \rho I.$$

Because $Q = TT^*$, its domain being all of H_A implies in particular $\text{dom } T^* = H_A$. For every $h \in H_A$,

$$\|T^*h\|^2 = (TT^*h \mid h)_A = (Qh \mid h)_A \leq \rho \|h\|^2.$$

Thus T^* is bounded by $\sqrt{\rho}$. Its adjoint $T = (T^*)^*$ is bounded with the same norm. Recalling the definitions of T, T^* proves (5).

If $\sigma(S) = \emptyset$, the same argument applies for every $\lambda > 0$, so (10) holds with $\rho = 0$ and $Q = 0$. This verifies the stated convention and completes the proof. $\square$

Verification and stress tests

The proof is analytic and uses no computation. Six focused audits were performed before promotion.

1. The quotient issue caused by a possible kernel of A disappears in (8): one solves in $\text{dom } A$ and then applies A . No induced inverse is assumed.
2. The product aligned with BC is TT^* , not T^*T , as the order in (7) confirms.
3. The source proves the mutual-adjoint identities under exactly the standing bounded-spectrum hypothesis, so TT^* is legitimately self-adjoint.
4. A possible spectral atom at λ is automatically annihilated in $d\mu_{(\lambda-Q)_u} = |\lambda - t|^2 d\mu_u$.
5. Lemma 1 uses pointwise finite potential and a covering argument, so it also rules out singular Cantor-type measures; it is not an unjustified absolute-continuity argument.
6. Bounded spectral support is invoked only after Q is known self-adjoint. At that point the spectral theorem makes its domain all of H_A , which then forces T^* everywhere defined.

The script `code/crop_open_problem.py` reproducibly creates the two source crops from pages rendered at 180 dpi. It is evidence preparation, not a proof verifier.

Scope, novelty, and human review

Theorem 1 fully answers the source’s boundedness question under the hypotheses in which the auxiliary pair was introduced, and recovers the same spectral-radius bound previously recorded only in the Banach setting. It does not remove Theorem 3.3’s assumption that $\sigma(BC|_{\text{dom } A})$ is bounded, nor does it put a norm on the ambient anti-dual-pair spaces.

A bounded search through August 11, 2026 covered the run’s four lightweight indexes, the local arXiv source corpus, the final 2021 journal version, and official-arXiv queries using the identifier, exact title and open sentence, author names, auxiliary operators, anti-dual pairs, boundedness, and spectral radius of BC . The final journal version still states the question as open, and no later explicit answer was found. This is not a priority claim.

Human review should focus on (7), the use of source resolvent surjectivity in (8), and the outer-measure version of the $5r$ -covering argument in Lemma 1. The empty-spectrum edge case should also be checked.

References

- [1] Zs. Tarcsay and T. Titkos, *Operators on anti-dual pairs: Generalized Krein–von Neumann extension*, Math. Nachr. **294** (2021), no. 9, 1821–1838; arXiv:1810.02619. Especially Theorem 3.3 and the following paragraph.
- [2] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, revised and enlarged edition, Academic Press, 1980.
- [3] P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces*, Cambridge Studies in Advanced Mathematics 44, Cambridge University Press, 1995.